

JAYDEN CHAN

☎ 647-616-5987 ✉ chanjayden31@gmail.com  [linkedin.com/in/Jayden-Chan](https://www.linkedin.com/in/Jayden-Chan)  github.com/JaydenChan2

Education

University of Waterloo

Bachelor of Computer Science — Double Degree Program

May 2030

Waterloo, Ontario

Wilfrid Laurier University

Bachelor of Business Administration — Double Degree Program

May 2030

Waterloo, Ontario

Technical Skills

Languages: Python, C/C++, Java, Javascript, Racket, HTML/CSS, SQL

Frameworks & Libraries: React, FastAPI, Flask, TensorFlow, OpenCV, MediaPipe, NumPy, Pandas, Librosa, Matplotlib

Tools & Platforms: Git/GitHub, Linux/Bash, Google Gemini API, ElevenLabs, Backboard.io, FFmpeg, SceneKit, Metal, RESTful APIs

Concepts: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Computer Vision, Digital Signal Processing (DSP), Technical Analysis, Agile Development

Experience

Software Developer

July 2024 – October 2024

FuturIQ Inc.

Brampton, Ontario

- **Increased user engagement** with AI interfaces by developing an **interactive 3D digital human** in **Unity** using **MetaHuman Creator** to simulate realistic, human-like interactions.
- **Streamlined social media workflows** by engineering a **customized chatbot** in **Python** designed to act as an automated advisor for content optimization.
- **Enhanced web accessibility** for AI tools by building an **interactive HTML/CSS site** featuring a **beginner-friendly GUI** and seamless **chatbot integration**.

Artificial Intelligence Instructor

September 2024 – June 2025

Dufferin-Peel Catholic District School Board

Brampton, Ontario

- **Architected and delivered technical curriculum** centered on deep learning architectures, guiding the implementation of **Convolutional Neural Networks (CNNs)** and **Recurrent Neural Networks (RNNs)** using **TensorFlow** and **Keras**.
- **Engineered end-to-end computer vision pipelines** utilizing **OpenCV** and **MediaPipe** for real-time feature extraction and object detection, emphasizing the optimization of model inference and data augmentation techniques.
- **Facilitated hands-on development workshops** for deploying AI models via **RESTful APIs**, mentoring students on the transition from local **Python** environments to scalable, integrated web applications.

Projects

Lynx

March 2026

- **Engineered an autonomous AI fundraising platform** matching startup IP with institutional capital, achieving an estimated **85% mandate-alignment accuracy** by developing a full-stack application using **React** and **FastAPI**.
- **Architected a real-time adversarial VC simulator** maintaining **sub-800ms conversational latency** by integrating the **Google Gemini API** for dynamic persona generation and **ElevenLabs** for synthetic audio.
- **Built a deep document ingestion pipeline** parsing **50+ pages of dense technical IP** into a structured knowledge graph by utilizing **Backboard.io** for RAG and multi-session memory tool-calling.

Music Sheet Converter

February 2026 – Present

- **Automated the music transcription process** from multi-format audio/video sources by engineering an **AI-driven system** that outputs high-fidelity **sheet music** and **instrument tabs**.
- **Improved translation accuracy** of complex tutorials into **playable notation** by implementing a **multimodal feature** that synchronizes visual cues with audio instruction.
- **Standardized audio output formats** including **MIDI** and **PDF** by implementing **advanced audio processing** to identify precise pitch and rhythmic intervals.

AI Facial Geometry Analysis

December 2025

- **Achieved real-time 3D geometry analysis** of facial structures by building a **computer vision application** in **Python** utilizing **MediaPipe** to track **468 facial landmarks**.
- **Quantified structural facial aesthetics** by developing a **mathematical engine** using **NumPy** to calculate metrics such as **facial symmetry**, **canthal tilt**, and **golden ratio alignment**.
- **Automated objective harmony measurements** by implementing **vector calculus algorithms** to analyze spatial proportions and facial ratios.